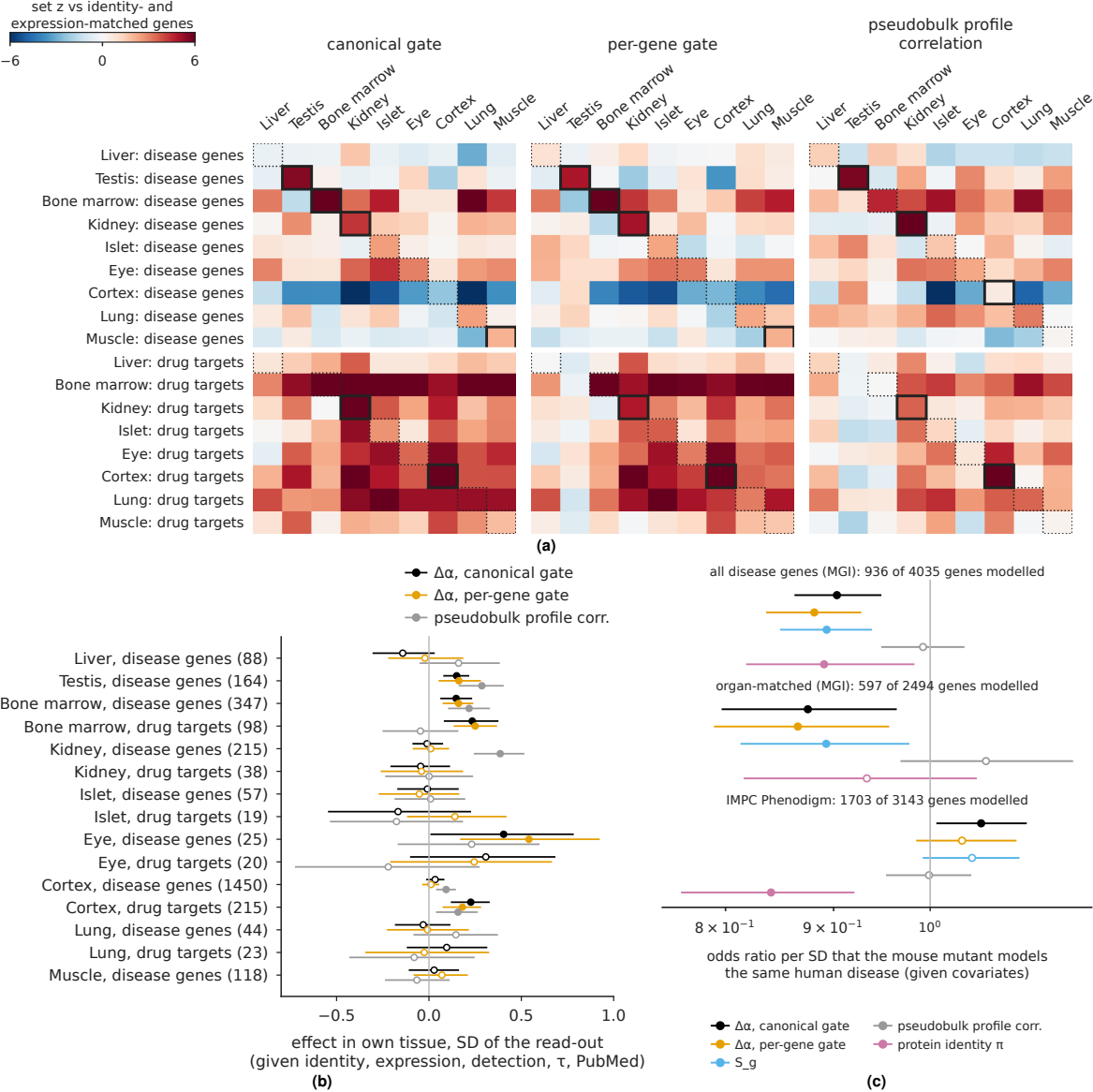

set z vs identity- and expression-matched genes
−6
0
6
canonical gate
per-gene gate
pseudobulk profile correlation
Liver
Testis
Bone marrow
Kidney
Islet
Eye
Cortex
Lung
Muscle
Liver: disease genes
Testis: disease genes
Bone marrow: disease genes
Kidney: disease genes
Islet: disease genes
Eye: disease genes
Cortex: disease genes
Lung: disease genes
Muscle: disease genes
Liver: drug targets
Bone marrow: drug targets
Kidney: drug targets
Islet: drug targets
Eye: drug targets
Cortex: drug targets
Lung: drug targets
Muscle: drug targets
(a)
Δα, canonical gate
Δα, per-gene gate
pseudobulk profile corr.
Liver, disease genes (88)
Testis, disease genes (164)
Bone marrow, disease genes (347)
Bone marrow, drug targets (98)
Kidney, disease genes (215)
Kidney, drug targets (38)
Islet, disease genes (57)
Islet, drug targets (19)
Eye, disease genes (25)
Eye, drug targets (20)
Cortex, disease genes (1450)
Cortex, drug targets (215)
Lung, disease genes (44)
Lung, drug targets (23)
Muscle, disease genes (118)
−0.5
0.0
0.5
1.0
effect in own tissue, SD of the read-out
(given identity, expression, detection, τ, PubMed)
(b)
all disease genes (MGI): 936 of 4035 genes modelled
organ-matched (MGI): 597 of 2494 genes modelled
IMPC Phenodigm: 1703 of 3143 genes modelled
8×10^{-1}
9×10^{-1}
10^{0}
odds ratio per SD that the mouse mutant models
the same human disease (given covariates)
Δα, canonical gate
Δα, per-gene gate
S_g
pseudobulk profile corr.
protein identity π
(c)